HARD · GEOMETRY AND TRIGONOMETRY · ANSWER KEY

# Right triangles and trigonometry

03

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ANSWER KEY · SAT QUESTION BANK

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## Q1 338

The correct answer is 338. The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. In triangle $XYZ$, it's given that angle $Y$ is a right angle. Thus, $\bar{XY}$ is the leg opposite of angle $Z$ and $\bar{YZ}$ is the leg adjacent to angle $Z$. It follows that $\tan Z = \frac{XY}{YZ}$. It's also given that the measure of angle $Z$ is $33°$ and the length of $\bar{YZ}$ is 26 units. Substituting $33°$ for $Z$ and 26 for $YZ$ in the equation $\tan Z = \frac{XY}{YZ}$ yields $\tan 33° = \frac{XY}{26}$. Multiplying each side of this equation by 26 yields $26\tan 33° = XY$. Therefore, the length of $\bar{XY}$ is $26\tan 33°$. The area of a triangle is half the product of the lengths of its legs. Since the length of $\bar{YZ}$ is 26 and the length of $\bar{XY}$ is $26\tan 33°$, it follows that the area of triangle $XYZ$ is $\frac{1}{2}2626\tan 33°$ square units, or $338\tan 33°$ square units. It's given that the area, in square units, of triangle $XYZ$ can be represented by the expression $k\tan 33°$, where $k$ is a constant. Therefore, 338 is the value of $k$.

## Q2

D

D. $\frac{3}{\sqrt{3}}$

Choice D is correct. The tangent of a nonright angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. Using that definition for tangent, in a right triangle with legs that have lengths a and b, the tangent of one acute angle is $\frac{a}{b}$ and the tangent for the other acute angle is $\frac{b}{a}$. It follows that the tangents of the acute angles in a right triangle are reciprocals of each other. Therefore, the tangent of the other acute angle in the given triangle is the reciprocal of $\frac{\sqrt{3}}{3}$ or $\frac{3}{\sqrt{3}}$.

Choice A is incorrect and may result from assuming that the tangent of the other acute angle is the negative of the tangent of the angle described. Choice B is incorrect and may result from assuming that the tangent of the other acute angle is the negative of the reciprocal of the tangent of the angle described. Choice C is incorrect and may result from interpreting the tangent of the other acute angle as equal to the tangent of the angle described.

## Q3 D

D. $\frac{18}{\sin Q}$

Choice D is correct. The figure shows that triangle $QRS$ is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side $RS$ is $18$. Since only the equation $\sin Q = \frac{RS}{QS}$ contains $RS$, an expression representing $QS$ can be found by substituting $18$ for $RS$ in this equation, which yields $\sin Q = \frac{18}{QS}$. Multiplying both sides of this equation by $QS$ yields $QS \cdot \sin Q = 18$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{18}{\sin Q}$. Therefore, the expression $\frac{18}{\sin Q}$ represents the length of $QS$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

## Q4 104

The correct answer is $104$. An equilateral triangle is a triangle in which all three sides have the same length and all three angles have a measure of $60°$. The height of the triangle, $k\sqrt{3}$, is the length of the altitude from one vertex. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the $60°$ angle in each 30-60-90 right triangle. Since the altitude has a length of $k\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each $30°$ angle has a length of $k$ and each hypotenuse has a length of $2k$. In this case, the hypotenuse of each 30-60-90 right triangle is a side of the equilateral triangle; therefore, each side length of the equilateral triangle is $2k$. The perimeter of a triangle is the sum of the lengths of each side. It's given that the perimeter of the equilateral triangle is $624$; therefore, $2k + 2k + 2k = 624$, or $6k = 624$. Dividing both sides of this equation by $6$ yields $k = 104$.

## Q5

B

**B.** 168

Choice B is correct. It's given that angle $Z$ in triangle $XYZ$ is a right angle. Thus, side $YZ$ is the leg opposite angle $X$ and side $XZ$ is the leg adjacent to angle $X$. The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. It follows that $\tan X = \frac{YZ}{XZ}$. It's given that $\tan X = \frac{12}{35}$ and the length of side $YZ$ is 24 units. Substituting $\frac{12}{35}$ for $\tan X$ and 24 for $YZ$ in the equation $\tan X = \frac{YZ}{XZ}$ yields $\frac{12}{35} = \frac{24}{XZ}$. Multiplying both sides of this equation by $35XZ$ yields $12XZ = 2435$, or $12XZ = 840$. Dividing both sides of this equation by 12 yields $XZ = 70$. The length $XY$ can be calculated using the Pythagorean theorem, which states that if a right triangle has legs with lengths of $a$ and $b$ and a hypotenuse with length $c$, then $a^2 + b^2 = c^2$. Substituting 70 for $a$ and 24 for $b$ in this equation yields $70^2 + 24^2 = c^2$, or $5{,}476 = c^2$. Taking the square root of both sides of this equation yields $\pm 74 = c$. Since the length of the hypotenuse must be positive, $74 = c$. Therefore, the length of $XY$ is 74 units. The perimeter of a triangle is the sum of the lengths of all sides. Thus, $74 + 70 + 24$ units, or 168 units, is the perimeter of triangle $XYZ$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the perimeter, in units, for a right triangle where the length of side $YZ$ is 12 units, not 24 units.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q6

.9811

Accept also: 52/53

The correct answer is $\frac{52}{53}$. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. The length of the hypotenuse of the right triangle shown is 53. It's given that $RS = \sqrt{105}$. Therefore, the length of one of the legs of the triangle shown is $\sqrt{105}$. Let $x$ represent $TS$, the length of the other leg of the triangle shown. Therefore, $53^2 = \left(\sqrt{105}\right)^2 + x^2$, or $2{,}809 = 105 + x^2$. Subtracting $105$ from both sides of this equation yields $2{,}704 = x^2$. Taking the positive square root of both sides of this equation yields $52 = x$. Therefore, $TS$, the length of the other leg of the triangle shown, is $52$. The sine of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the hypotenuse. The length of the leg opposite angle $R$ is $52$, and the length of the hypotenuse is $53$. Therefore, the value of $\sin R$ is $\frac{52}{53}$. Note that 52/53 or .9811 are examples of ways to enter a correct answer.

# Q7

A

**A.** 20

Choice A is correct. When a square is inscribed in a circle, a diagonal of the square is a diameter of the circle. It's given that a square is inscribed in a circle and the length of a radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. Therefore, the length of a diameter of the circle is $2\frac{20\sqrt{2}}{2}$ inches, or $20\sqrt{2}$ inches. It follows that the length of a diagonal of the square is $20\sqrt{2}$ inches. A diagonal of a square separates the square into two right triangles in which the legs are the sides of the square and the hypotenuse is a diagonal. Since a square has 4 congruent sides, each of these two right triangles has congruent legs and a hypotenuse of length $20\sqrt{2}$ inches. Since each of these two right triangles has congruent legs, they are both 45-45-90 triangles. In a 45-45-90 triangle, the length of the hypotenuse is $\sqrt{2}$ times the length of a leg. Let $s$ represent the length of a leg of one of these 45-45-90 triangles. It follows that $20\sqrt{2} = \sqrt{2}s$. Dividing both sides of this equation by $\sqrt{2}$ yields $20 = s$. Therefore, the length of a leg of one of these 45-45-90 triangles is 20 inches. Since the legs of these two 45-45-90 triangles are the sides of the square, it follows that the side length of the square is 20 inches.

Choice B is incorrect. This is the length of a radius, in inches, of the circle.

Choice C is incorrect. This is the length of a diameter, in inches, of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q8

24

The correct answer is 24. The sine of an acute angle in a right triangle is equal to the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the sine of angle B, or $\sin(B)$, is equal to the ratio of the length of side $\overline{AC}$ to the length of side $\overline{AB}$. It's given that the length of side $\overline{AB}$ is 26 and that $\sin(B) = \frac{5}{13}$. Therefore, $\frac{5}{13} = \frac{AB}{26}$. Multiplying both sides of this equation by 26 yields $AC = 10$.

By the Pythagorean Theorem, the relationship between the lengths of the sides of triangle ABC is as follows: $26^2 = 10^2 + BC^2$, or $676 = 100 + BC^2$. Subtracting 100 from both sides of $676 = 100 + BC^2$ yields $576 = BC^2$. Taking the square root of both sides of $576 = BC^2$ yields $24 = BC$.

## Q9

.7241

Accept also: 21/29

The correct answer is $\frac{21}{29}$. It's given that triangle $ABC$ is similar to triangle $DEF$, where angle $A$ corresponds to angle $D$ and angles $C$ and $F$ are right angles. In similar triangles, the tangents of corresponding angles are equal. Therefore, if $\tan A = \frac{21}{20}$, then $\tan D = \frac{21}{20}$. In a right triangle, the tangent of an acute angle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. Therefore, in triangle $DEF$, if $\tan D = \frac{21}{20}$, the ratio of the length of $\bar{EF}$ to the length of $\bar{DF}$ is $\frac{21}{20}$. If the lengths of $\bar{EF}$ and $\bar{DF}$ are 21 and 20, respectively, then the ratio of the length of $\bar{EF}$ to the length of $\bar{DF}$ is $\frac{21}{20}$. In a right triangle, the sine of an acute angle is the ratio of the length of the leg opposite the angle to the length of the hypotenuse. Therefore, the value of $\sin D$ is the ratio of the length of $\bar{EF}$ to the length of $\bar{DE}$. The length of $\bar{DE}$ can be calculated using the Pythagorean theorem, which states that if the lengths of the legs of a right triangle are $a$ and $b$ and the length of the hypotenuse is $c$, then $a^2 + b^2 = c^2$. Therefore, if the lengths of $\bar{EF}$ and $\bar{DF}$ are 21 and 20, respectively, then $21^2 + 20^2 = DE^2$, or $841 = DE^2$. Taking the positive square root of both sides of this equation yields $29 = DE$. Therefore, if the lengths of $\bar{EF}$ and $\bar{DF}$ are 21 and 20, respectively, then the length of $\bar{DE}$ is 29 and the ratio of the length of $\bar{EF}$ to the length of $\bar{DE}$ is $\frac{21}{29}$. Thus, if $\tan A = \frac{21}{20}$, the value of $\sin D$ is $\frac{21}{29}$. Note that 21/29, .7241, and 0.724 are examples of ways to enter a correct answer.

# Q10 B

**B.** $\frac{5}{12}$

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Choice B is correct. It's given that right triangle $RST$ is similar to triangle $UVW$, where $S$ corresponds to $V$ and $T$ corresponds to $W$. It's given that the side lengths of the right triangle $RST$ are $RS = 20$, $ST = 48$, and $TR = 52$. Corresponding angles in similar triangles are equal. It follows that the measure of angle $T$ is equal to the measure of angle $W$. The hypotenuse of a right triangle is the longest side. It follows that the hypotenuse of triangle $RST$ is side $TR$. The hypotenuse of a right triangle is the side opposite the right angle. Therefore, angle $S$ is a right angle. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle $T$ is side $ST$. The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle $T$ is side $RS$. The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore, $\tan T = \frac{RS}{ST}$. Substituting 20 for $RS$ and 48 for $ST$ in this equation yields $\tan T = \frac{20}{48}$, or $\tan T = \frac{5}{12}$. The tangents of two acute angles with equal measures are equal. Since the measure of angle $T$ is equal to the measure of angle $W$, it follows that $\tan T = \tan W$. Substituting $\frac{5}{12}$ for $\tan T$ in this equation yields $\frac{5}{12} = \tan W$. Therefore, the value of $\tan W$ is $\frac{5}{12}$.

Choice A is incorrect. This is the value of $\sin W$.

Choice C is incorrect. This is the value of $\cos W$.

Choice D is incorrect. This is the value of $\frac{1}{\tan W}$.

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### Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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